

Mohammad Reza Aminian

Curriculum Vitae

The University of Chicago, Booth School of Business

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5807 S Woodlawn Ave, Chicago, IL 60637

Education

2021 - Present **Ph.D. in Business**, *The University of Chicago, Booth School of Business, Chicago, IL, USA*

Major: Management Sciences/Operations Management

Minor: Economics

Advisor: Prof. Rad Niazadeh

Expected Graduation Date: June 2027

2025 **M.B.A. in Business**, *The University of Chicago, Booth School of Business, Chicago, IL, USA*

2018 - 2021 **B.S. in Computer Science**, *Sharif University of Technology, Tehran, Iran.*

2016 - 2021 **B.S. in Electrical Engineering**, *Sharif University of Technology, Tehran, Iran.*

Research Interests

Socially-Responsible & Behaviorally-Aware Operation, EdTech, Algorithms, Learning, ML, RL, Optimization, Game Theory, Mechanism/Information Design

Research

Fundraising for Education: Optimizing Crowdfunding Platforms

Design dynamic assortment policies for crowdfunding platforms to prevent “project starvation” and maximize the value of fully funded campaigns.

with Rad Niazadeh (Chicago Booth)

Optimal Bayesian Online Allocation of Reusable Resources

Generalize classical online resource allocation theory to the reusable setting by developing a time-invariant admission policy that guarantees ex-post feasibility and matches the information-theoretic tight bounds of non-reusable models. [SSRN: 6384158](#)

with Rad Niazadeh (Chicago Booth), Pranav Nuti (Chicago Booth)

Stationary Online Contention Resolution Schemes

Create the S-OCRS framework to unify the design of online selection algorithms, achieving optimal selectability and resolving open problems across various feasibility environments, like matchings and matroids. [SSRN: 6384199](#)

with Rad Niazadeh (Chicago Booth), Pranav Nuti (Chicago Booth)

Accepted at EC 2026

Online Job Selection: Reward Rate vs. Remaining Value (formerly Real-Time Personalized Order Holding)

Introduce an online resource allocation framework and index-based policies with theoretical guarantees to optimize the trade-off between keeping an ongoing job to harvest its remaining value vs. preempting it to serve a new request in limited capacity systems. [SSRN: 4644495](#)

with Will Ma (Columbia GSB), Linwei Xin (Chicago Booth)

Accepted at Management Science

• **Featured in Chicago Booth Review**

Markovian Search with Ex-Ante Constraints: Theory and Applications to Socially Aware Algorithmic Hiring

Design socially-responsible sequential search framework for algorithmic hiring and resource allocation. Developed dual-adjusted index policies and primal-dual algorithms to satisfy ex-ante constraints efficiently. [SSRN: 4347447](#)

with Vahideh Manshadi (Yale SOM), Rad Niazadeh (Chicago Booth)

Accepted at Management Science, EAAMO 2024

- **Featured in Chicago Booth Review**
- **Third Place, IBM Service Science Best Student Paper Award, INFORMS 2023**

Teaching

Operations Management (Executive MBA), T.A., Chicago Booth, Winter 2024

Managerial Decision Modelling (Undergrad, MiM, MBA), T.A., Chicago Booth, Spring: 2023, 2024, Fall: 2024, 2025

Linear Programmig (Ph.D.), T.A., Chicago Booth, Fall: 2022, 2023

Math Olympiad Teacher and Tutor

Work

2019 - 2021 **Research Assistant at the *Sharif University of Technology, Tehran, Iran.***
Graph Auto-Encoder with capability of reconstructing any weighted adjacency matrix
Supervisor: Prof. Saber Saleh Kaleybar

2017 **Research Assistant at *Ambient Intelligence Research Lab (AIR Lab), Sharif University of Technology, Tehran, Iran.***
Signal Processing on the EEG data
Supervisor: Prof. Hamid K. Aghajan

Presentations

Responsible Decision Making in Operations: Algorithms for Social Impact

Chicago OPS Day 2026, Poster, Chicago, IL, USA

Rising Stars Workshop 2026, Ann Arbor, MI, USA

Stationary Online Contention Resolution Schemes:

INFORMS 2025, Atlanta, GA, USA

Purdue Operations Conference 2025, West Lafayette, IN, USA

RMP 2025, New York, NY, USA

EC 2025, Poster, Stanford, CA, USA

INFORMS 2024, Seattle, WA, USA

Online Job Selection: Reward Rate vs. Remaining Value (formerly Real-Time Personalized Order Holding):

RMP 2024, Los Angeles, CA, USA

Chicago OPS Day 2024, Poster, Chicago, IL, USA

INFORMS 2023, Phoenix, AZ, USA

ChicagoBooth Operations Day 2023, Chicago, IL, USA

MOPTA 2023, Bethlehem, PA, USA

MIW 2023, Virtual

Markovian Search with Socially Aware Constraints:

APS 2025, Atlanta, GA, USA

Purdue Operations Conference 2024, West Lafayette, IN, USA

EC 2024, Poster, New Haven, CT, USA

MSOM 2024, Minneapolis, MN, USA

FORC 2024, Cambridge, MA, USA

Chicago OPS Day 2024, Poster, Chicago, IL, USA

Midwest Workshop on Game Theory 2024, Poster, Chicago, IL, USA

Columbia Workshop on Fairness 2023, Poster, NYC, NY, USA

INFORMS 2023, Phoenix, AZ, USA

EC 2023, Poster, Virtual

MIW 2023, Virtual

NYC OPS Day 2023, Poster, NYC, NY, USA

INFORMS 2022, Indianapolis, IN, USA

Service

Journal Reviewing	Management Science, Operations Research, M&SOM
Conference Reviewing	MSOM, International Symposium on Algorithms and Computation (ISAAC)

Selected Honors and Awards (since 2012)

2023	Third Place, IBM Service Science Best Student Paper Award, <i>INFORMS</i>
2021 - 2023	K.D.M. Fellowship Award, <i>The University of Chicago, Booth School of Business</i>
2021 - 2027	Booth Doctoral Fellowship Award, <i>The University of Chicago, Booth School of Business</i>
2021	Honorary admission for Master of Business Administration, <i>Sharif University of Technology, Tehran, Iran</i>
2019	Top Second Prize Winner at International Mathematics Competition for university students, <i>Bulgaria</i>
2016	Honorary admission for Bachelor of Electrical Engineering, <i>Sharif University of Technology, Tehran, Iran</i>
2016	Silver medalist at 57th International Mathematical Olympiad (IMO), <i>Hong Kong</i>
2015	Gold medalist at 33rd National Mathematical Olympiad, <i>Iran</i>
2014	Silver medalist at International Mathematics Competition (KIMC), <i>South Korea</i>
2014	Second Runner up on both Team Prize and Group Prize at International Mathematics Competition (KIMC), <i>South Korea</i>
2014	Silver medalist at 32nd National Mathematical Olympiad, <i>Iran</i>
2014	Gold medalist at 15th National Mathematics Competition, <i>Iran</i>
2013	Gold medalist at 14th National Mathematics Competition, <i>Iran</i>
2013	Gold medalist at 3rd Mathematical Olympiad of Iranian Teenagers, <i>Iran</i>
2012	Merit medalist at International Mathematics Competition (TAIMC), <i>Taiwan</i>

2012 **Gold medalist at 13th National Mathematics Competition, Iran**

Languages

Human **Persian:** Native, **English:** Advanced, **Arabic:** Intermediate, **Turkish:** Intermediate, **Japanese:** Elementary

Computer Python, Matlab, R, Java, C++, C, $\text{\LaTeX}$

Extracurricular Activities

Anime, Swimming, Cycling, Archery, (Table) Tennis, Badminton, Hiking, Nature-tourism

Selected Courses

Graduate Approximate Dynamic Programming, Market Design, Large Scale Optimization, Learning and Game Theory, Supply Chain, Dynamic Programming, Econometrics 2 and 3, Online Optimization 1 and 2, Approximation Algorithms, Stochastic Processes, Linear Programming, Advanced Microeconomics Analysis, Microeconomics 2 and 3, Graph Representation Learning, Deep Learning, Machine Learning, Causal Inference, Distributed Systems, Data Networks, Principles of Entrepreneurship, Convex Optimization 2, Queuing Theory (Both Sharif and Booth)

Under Graduate Artificial Intelligence, Introduction to Machine Learning, Graph Theory, Linear Algebra, Design Algorithms, Convex Optimization 1, Stochastic Processes, Probability and Statistics, Data Structures, Automata and Languages Theory, Numerical Analysis 1 and 2, Math Logic, Introduction and Advanced Programming, Math Analysis, Discrete Math, Differential Equations, Principles of Economy, Combinatorics